

Microtubule Theory of Sensory Transduction

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It is proposed that ciliary microtubules are actively involved in the reception and transduction of sensory information from the distal portion of the dendrite to the basal body area of the receptor cell. Environmental stimuli may cause stimulus-specific conformational changes in the tubulin protein molecule. This characteristic change may then be propagated along the tubulin polymer filaments. Once this coded information has reached the cell body, classical ion flux mechanisms are thought to further amplify the stimulus energy and transmit its patterned information (generator potential) to the action potential generating sites of the cell. Morphological and biochemical evidence for this point of view is presented.

Current theories on odor perception assume the presence of several to many different receptor sites (Amoore, 1970; Schneider, 1969), which are tacitly thought to be located in the cell membrane. Reviewing some of the published accounts on the ultrastructure of vertebrate and invertebrate olfactory receptors (De Lorenzo, 1963; Frish, 1967; Moulton & Beidler, 1967; Bannister, 1965; Ernst, 1969; Ghiradella, Case & Cronshaw, 1968 *a,b,c*) one finds a striking similarity between a great number of organelles and cellular arrangements in this type of receptor and in the receptors of other sensory systems, such as the acoustico-lateralis system (Engström & Wersäll, 1958; Flock, 1967, 1971) and common photoreceptors (Eakin, 1963; Tomita, 1970). Based on these morphological similarities, for which Vinnikov (1970) proposed an evolutionary model, and on new biochemical information, it is proposed that the ciliary and microtubular apparatus in sensory cells receive the environmental information and transmit it via propagated conformational changes in microtubule protein. The argument will be presented that microtubules are active, functional units in the reception and transduction of such stimuli.

The receptor cells of the sensory systems mentioned above are ciliary in nature. Each cilium contains the 9+2 arrangement of microtubules that is

characteristic of motile cilia and flagella (Fawcett, 1961, 1966) and is likewise derived from a basal body. A rootlet system and cellular microtubules are often observed. Many mitochondria are present in the basal body area, which appears hyaline except for the presence of cellular microtubules, in contrast with the surrounding granular cytoplasm; the nucleus is round, central and surrounded by endoplasmic reticulum. Regardless of whether it is a primary receptor or a secondary receptor (the latter are connected to more central parts of the nervous system by separate neurons), the sensory cell is surrounded by supporting cells and forms specialized junctions with them, such as tight and gap junctions (Brightman & Reese, 1969); desmosomes can be observed in many places. Basal cells underly the sensory epithelium. While other receptor systems could be mentioned as well, we will concentrate on the mechanoreceptor (hair cell) of the acoustico-lateralis system, the chemoreceptor of the olfactory system and the photoreceptor of the vertebrate lateral eye.

Differences in structure of olfactory receptor cells and mechanoreceptor cells are in some cases seen in the peripheral arrangement of microtubules in the olfactory cell—each cilium can divide into many branches that each contain one microtubule from the original $9+2$ structure (Ghiradella *et al.*, 1968 *a,b,c*; Ernst, 1969) whereas, in the mechanoreceptor cell, the cilium remains always undivided (Flock, 1967). An undivided olfactory cilium occurs in the sensilla trichodea of male *Saturniid* moths in specialized pheromone receptor cells; the generalist olfactory receptor cells in the sensilla basiconica have branched cilia (Schneider, Lacher & Kaissling, 1964; see also Slifer, Prestage & Beams, 1959, for grasshopper). In photoreceptors, an undivided cilium is seen with, in the distal segment, specialized structures in close proximity to the microtubules.

Lowenstein, Osborne & Wersäll, (1964) suggested that the kinocilium of the hair cells in the acoustico-lateralis system might act as a motile cilium in reverse; the idea being that cilia produce movement using chemical energy provided by the cell; and vice versa, that mechanoreceptor cilia undergo mechanical deformation caused by external forces and provide the cell with a pattern of chemical energy (which represents information). The cilium and its related structures are thought to function as a chemo-mechanical transducer system that can work both ways. The underlying principle of such a chemo-mechanical transducer can be demonstrated with a simple example: a polymer with multiple carboxyl groups has been synthesized, which shortened its chain when calcium was added. When the contracted chain was stretched mechanically, calcium was expelled (Kuhn, 1949). Another example, which applies directly to the arguments presented here, is found in the mechanosensitivity of motile cilia on the gill of the mussel, *Mytilus* (Thurm,

1968). From these studies one may conclude that sensitive mechanisms, i.e. reception of energy, developed in close functional relationship with the ciliary motor mechanism, i.e. production of energy.

It is generally accepted that the kinocilium is actively involved in the production of movement and that the energy and the control are supplied by centers in the cell body, perhaps the basal bodies (see for example Allen, 1971). Whether or not the reverse situation exists, however, is still open to debate. Some authors believe that cilia and microtubules in receptor cells merely function as rods to support the fragile cell protrusions of these cells (Porter, 1966) and have no "active" function. Recent investigations in Porter's laboratory have shown that microtubules are essential for stimulus transduction in an insect mechanoreceptor. Treatment with high doses of vinblastine or colchicine disassembled the microtubules in the dendritic end of the receptor cell of the campaniform sensillum of the cockroach leg. It also inhibited the neurophysiologically recorded response to a mechanical stimulus. The authors interpret the function of the 350–1000 microtubules in each cell to be a stiff translation rod, causing local membrane depolarization (Moran & Varela, 1971).

Though one cannot disprove this interpretation in the case of a mechanoreceptor, where active and passive transduction by microtubules cannot be distinguished, arguments will be presented now to demonstrate that microtubules may well represent active functional units involved in the reception, transformation and conduction of stimuli in general.

The beginning of an understanding of the structure and function of tubulin have emerged (Stephens, 1971). The fact that conformational changes in microtubule protein arrangement have been observed both in fixed and fresh material (Thomas, 1970; Cohen, Harrison & Stephens, 1971; Ventilla, Cantor & Shelanski, 1972), suggests that these changes occur naturally, perhaps under environmental influences. This suggestion became more convincing when it was found that oscillations of sperm flagella are probably not controlled by the cell membrane, but are properties of microtubules in the presence of ATP. Neither the sperm head, nor the membrane, nor the centriole and midpiece are necessary to produce flagellar waves in sperm tails (Lindemann & Rikmenspoel, 1972).

Microtubules consist of globular elements lined up in long strands of filaments; 12–13 filaments form the shaft of the microtubule which has a characteristic diameter of about 250 Å. The globular elements of neighboring filaments are in a staggered or semi-staggered configuration. Each globule is a coiled protein of which about one third is alpha-helical (Stephens, 1969). Changes in configuration can theoretically take place in several different ways within each globular protein and between the globules.

From what we know of the structure of the microtubule and of similar structures and models, we can conclude that propagation of local distortions is not only possible, but rather likely to take place in microtubules. In motile cilia and flagella, the distortion must originate at the base and must be propagated distally. It involves displacement of a tubule section. In sensory cilia the distortion must originate somewhere along the length of the tubules and may be propagated in either direction, or at least in a proximal direction, if special asymmetrical configurations act as intra-molecular "synapses". The distortion does not necessarily involve displacement of a tubule segment, but is more likely a sequence of conformational changes.

Thus, assuming that sensory microtubules are capable of this type of wave propagation, it is easily seen that both gross mechanical deformation and local chemical distortion can act as a stimulus to start a wave. For this reason it does not seem impossible that not only mechanoreception, but also chemoreception may take place on the microtubules themselves. Some rather complex alkaloids, such as colchicine, vinblastine, etc., are known to bind preferentially to tubulin, thereby disrupting their polymerization (Borisy & Taylor, 1967*a,b*). Direct tubulin reception leaves the cell membrane with perhaps a screening permeability function as is assumed in other cells in general. Other authors have already speculated that olfactory receptor cells may operate "by virtue of transmitted mechanical signals to the base of the cilium" (Lettvin & Gesteland, 1965).

As mentioned earlier, mechanoreceptors tend to favor the 9+2 arrangement, all the way up to the tip of the cilium, possibly to assure greater mechanical strength, whereas some chemoreceptors send out individual microtubules branching off somewhere distally from the basal bodies. This structure would indeed have advantages for a chemoreceptor where mechanical deformation is not to be feared. The branching would increase the molecule-catching capacity of each cell considerably, and thereby enhance the sensitivity of chemosensory detection. Since individual microtubules, as far as known, are built of identical, repeating globular units, we might expect that chemoreception can take place over the full length and diameter of the tubule.

There is both morphological (Behnke & Forer, 1967) and chemical evidence (Stephens, 1970; Jacobs & McVittie, 1970) for different classes of tubulins. Each of the outer doublet microtubules of flagella is composed of a distinct tubulin moiety. *A* and *B* tubulin, derived from *A* and *B* tubules, differ in isoelectric point, amino acid composition and tryptic peptide maps. They can be thermally fractionated (Stephens, 1970). Different tubulins appear in different animal species, as well as in different organs in the same species (Linck, 1970).

The structure of tubulins—as little as is presently known—leaves ample room to accommodate many different interaction sites with foreign molecules, such as odorants. Each odorant is thought to interact through weak bonds at one or more different sites to produce a pattern of conformational changes (see for instance Norris, 1971). Some odor molecules (*A*) may be able to interact at only one site in one way, others (*B*) may interact at many sites in many ways. Each combination of interactions will result in an odor wave pattern and represent a code for odor quality. Quantity is related to the number of sites and microtubules that are stimulated. The implications are that substance *A* is recognizable in low concentrations if there are many *A* sites; *B* may smell different in lower concentrations or be odorless until it interacts with enough different sites to be recognized as such. Kafka (1971) tested nearly 370 compounds for their neural response in the olfactory pit organs of the migratory locust and concluded that the number of available receptor sites (for a certain odor molecule) and the accessibility of the site determine the threshold sensitivity of the olfactory receptor for that particular molecule. Olfactory saturation may be determined, in part, by the properties of the most peripheral structures of the system, the interaction sites, which become temporarily filled up.

Once an odorant molecule has interacted with a site, a molecular refractory period can be postulated. Metabolic processes in the cell evidenced by the large concentration of mitochondria around the basal end of the cilia are likely to be involved in the restoration of the tubular “receptor potential”. Of course, refractory periods may also arise in more central parts of the olfactory system.

In all sensory systems, the signal patterns, which eventually give rise to the graded generator potentials at the proximal cell membrane, must already be represented in the original microtubular distortions as caused by environmental stimuli. It has been shown that a stimulus quantum can produce a discrete potential fluctuation across the receptor cell membrane. With increased stimulation the number of these small fluctuations increases and gives rise to the graded receptor potential (Adolph, 1964). It is possible that each quantal fluctuation at the proximal membrane represents one discrete conformational change in the microtubular apparatus.

Once the stimulus has reached the transition zone between the cilium and the cell apex (i.e., the basal body area) a different transduction mechanism most likely takes over to reach the afferent neuron. This is the electrical process which, according to most authors, is responsible for the entire conduction process from the primary conformational change, via local depolarization of the ciliary membrane, to the generator potential. A very attractive model for this process is being developed by Thurm (1970 and

personal communication) who has based it on the conductivity properties of epithelia in general. The high resistance barriers between sensory and supporting cells and the structure and function of specialized junction areas play a crucial role in his concept, which finds support by other authors (MacLeod, 1959; Reese & Brightman, 1970).

The idea that cilia contain the interaction sites of the olfactory process is not new (Hopkins, 1926); several authors have stressed the importance of cilia (Reese, 1965). In recent years, however, some investigators have argued against cilia. Their arguments are based on the recently discovered absence of cilia on the bipolar cells in the olfactory epithelium of the guitar fish, an elasmobranch (Reese & Brightman, 1970) and the absence of cilia in the vomeronasal organ of the turtle (Graziadei & Tucker, 1970). In the case of the vomeronasal organ, which responds in a way similar to an olfactory organ, up to 20 large centrioles were described independently by Bannister and by Altner for several amphibians, reptiles and mammals (in press and personal communication). The presence of centrioles weakens general arguments against ciliary participation, but further investigation of the physiology of the vomeronasal organ is clearly needed. The olfactory epithelium of the guitar fish shows bipolar cells alternating with so-called supporting cells, which carry several undivided cilia. Thurm's model (1970) stresses the essential involvement of supporting cells, and there is little reason here to exclude these cells from participation in the sensory process.

In the mechanoreceptor system a similar exception to the ciliary rule is found in the mammalian organ of Corti, where the kinocilium is reduced in ontogenetic development and absent in the adult. A centriole or basal body remains in its strategic position at the cell apex and a brush of stereocilia may act as a stiff translation rod for stimulus conduction to the basal body area in this mechanoreceptor.

A strong argument in favor of the ideas expressed here comes from Ottoson & Shepherd (1967), who calculated that a cylinder with the length and diameter of an olfactory cilium could initiate and conduct an effective electrical potential to the bipolar dendrite, only if special properties were assumed. Also, Beidler (1970) discussing mechanisms of taste receptor stimulation concluded that "a conformational change in the receptor molecule must produce an effect that is transmitted to areas of the cell distant from the receptor sites by means other than electrical depolarization associated with changes in ionic fluxes. The process by which such effects can be propagated is not yet known" (p. 61). The microtubular system may provide for such conduction properties. All presently available morphological and chemical evidence of microtubule structure supports this view.

The concept developed here explains primary olfactory receptor mechanisms in the same way commonly assumed membrane bound mechanisms do, but opens the possibility for direct involvement of microtubules. In this way it also provides a structural basis for many current theories of primary olfactory mechanisms. For instance, specific anosmias (Amoore, 1970) can be explained as caused by a genetic change in the amino acid sequence or stereometrical configuration of tubulin monomer. Schneider's (1969) concept of odor specialists which respond to only one odor (pheromone) and odor generalists which respond to a spectrum of odors may lead to an experimental test of the microtubule theory. The two types of olfactory receptor cells can be distinguished morphologically and electrophysiologically (Boeckh, 1969). When surgically separated, two different tubulins may be extracted; one which bonds with the pheromone and not with odorants in general, and the other should bond with different odorants in specific ratios. A few different tubulins may exist in the generalist receptor cells, which then explains how individual generalist cells respond with different, but cell-characteristic odor spectra (Gesteland, Lettvin, Pitts & Rojas, 1963; Lacher, 1964; Schneider *et al.*, 1964; Shibuya & Tucker, 1967).

The microtubule theory assumes that excitability to a variety of external stimuli is a property of all cells, but it is increased in receptor cells. These specialized cells generally utilize a cilium to receive and transmit the stimulus to the cell body. Responsiveness to more than one stimulus modality has been suggested by von Békésy (1964), Nott & Foster (1969), and Andersen (1970). Receptor specificity can be obtained by special receptor molecules which increase sensitivity for a specific class of stimuli, such as rhodopsin for photoreception. It may be irrelevant whether the receptor molecule is part of the membrane system or the microtubular system. In some cases the two may be intimately connected as was recently suggested by the work of Shelanski (1972), Stadler & Franke (1972) and Feit, Slusarek & Shelanski (1971).

The microtubule theory assumes that the adequate stimulus for the microtubule is distortion of tubulin configuration, which can be affected by any one source of environmental energy. Then specialized cellular structures, such as the different arrangements in which microtubules are found, adapt the receptor cell to the reception of different classes of stimuli, i.e. light energy, chemical bond energy and mechanical forces. It follows that the adequate stimulus for the receptor cell is more specialized than for the microtubule.

A parallel of this phenomenon can be found at the next higher level of organization. The receptor cells themselves, in turn, are often arranged in

epithelia of a special geometry or connected with specialized organs which insure that only specific parts of a stimulus class are perceived, for instance a band from a frequency spectrum. An example is the different arrangements in which hair cells occur. The adequate stimulus is probably the same for each hair cell regardless of where it is found: a mechanical distortion of the basal body of the kinocilium. But, in the cochlea, haircells are connected with the tectorial membrane for sound perception; in the crista ampullaris they are connected with the cupula of the semicircular canals for the perception of angular acceleration; in the maculae of the vestibular apparatus they are connected with the otoliths for the perception of linear acceleration, especially the gravitational force; and in the lateral line of fishes and amphibia they are again embedded in a cupula, but here for the detection of water displacements, etc.

The unique features of tubulin and microtubules may have been recognized early in evolution and they probably still exert an extraordinary influence in contemporary forms of life. The most primitive stages of perception steering behavior may be found in bacterial chemotaxis. Chemo-reception in bacteria affects their flagellar beat (Adler, 1965). Specific "anosmias", based on genetic mutations occur. Although bacterial flagella are different from cilia in Metazoa and Protozoa, similar principles of protein transduction may apply. The Precambrian evolution of spirochaete symbiosis with other primitive cells into eukaryotes (Margulis, 1970) provided the cells with ciliar motility, as well as sensitivity. In subsequent refinements some ciliates may have differentiated cilia of a type that may be exclusively sensory, possibly mechanosensory. This is the case in *Nassula*, among other contemporary ciliates, where there is a clear differentiation into motile and sensory cilia (Tucker, 1971).

In the highest evolutionary development, the mammalian brain, much of the tubulin material is found in the particulate fraction of brain extracts rather than in the soluble fraction (Shelanski, 1972; Stadler & Franke, 1972; Feit *et al.*, 1971) indicating close association with the membrane. It may not be too far to speculate that the biochemical machinery makes use of the reception and transduction properties of tubulin protein and incorporates them into the membrane system of mature nerve cells for communication purposes. The fact that microtubules in brain cells and axonal neurotubules chemically and morphologically show great similarities to ciliary and flagellar microtubules (Feit *et al.*, 1971) makes it even more pertinent to consider the possibility of microtubular involvement in chemical binding, information transmission, and intracellular communication.

The microtubule theory of sensory transduction can be tested with modern techniques of electron microscopy and autoradiography: the tubules, not

the outer membrane, ought to bind known odorants; crystallography: isolation and structural determination of molecules such as odorants and pheromones bound to tubulin protein; protein chemistry: slight differences in primary structures of tubulins sensitive to different odorants without any changes in the corresponding membrane lipoproteins. Genetic mutants unable to perceive odorants (for instance organisms unable to taste PTC) ought to have altered tubulins rather than altered membrane lipoproteins. Circular dichroism: tubulin protein from olfactory cilia *in vitro* must show conformational changes when its specific sex pheromone is added. It has already been shown that pig brain tubulin responds to pH, temperature and colchicine binding in this way (Ventilla *et al.*, 1972). Molecular "lesions" of tubules conducted in a mechanoreceptor system cannot show the difference between a passive and an active transduction role of microtubules. However, similar experiments on photoreceptors or chemoreceptors may give positive evidence for the validity of this theory, which presents a unifying concept of information transmission in sensory receptor cells.

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